

DS	Session	Hashing	Question Information	Level 1 Challenge 94
Question description The students of college <i>XYZ</i> are getting jealous of the students of college <i>ABC</i>. <i>ABC</i> managed to beat <i>XYZ</i> in all the sports and games events. The main strength of the students of <i>ABC</i> is their unity. The students of <i>XYZ</i> decide to destroy this unity. The geeks of <i>XYZ</i> prepared a special kind of perfume. Anyone who inhales this perfume becomes extremely violent. The students of <i>XYZ</i> somehow manage to spread this perfume throughout <i>ABC</i>'s campus atmosphere. There are N boys $(1, 2, 3, \dots, N)$ and N girls $(1, 2, 3, \dots, N)$ in <i>ABC</i> college. Each boy has a crush on a single girl and each girl has a crush on a single boy. Since the perfume has been inhaled by each and every student of <i>ABC</i> college, every student decides to beat up his/her crush's crush, ie. , if boy x has a crush on girl y and girl y has a crush on boy z, x will beat z up, provided, of course, if x and z is not the same person. The doctor of <i>ABC</i> college foresees this situation. He cannot stop so many people from beating each other up, however, he can be prepared for the worst-case patient(s). The worst-case patient(s) will be the patient(s) who get(s) beaten up by the maximum number of students. The doctor comes to you for help. He has 2 questions for you : 1. What is the number of beatings received by the worst-case patient(s) ? 2. What is the total number of pairs of students who ended up beating up each other ? Constraints : $1 \leq T \leq 10$ $1 \leq N \leq 10^5$ Input : The first line comprises of T, the number of test cases. Each test case comprises of 3 lines. The first line consists of N. The next line consists of N space separated natural numbers between 1 and N inclusive such that the ith number denotes the the crush of boy i. The next line consists of N space separated natural numbers between 1 and N inclusive such that the ith number denotes the the crush of girl i. Output : For every test case, on a new line, print two space separated integers, the answer to doctor's question 1 followed by answer to doctor's question 2.				

✓ Logical Test Cases

Test Case 1

INPUT (STDIN)

```
2
3
2 2 1
3 2 1
4
2 3 4 1
2 3 4 1
```

EXPECTED OUTPUT

```
1 0
1 4
```

Test Case 2

INPUT (STDIN)

```
2
4
4 2 1 3
4 2 1 2
3
2 4 1
2 4 1
```

EXPECTED OUTPUT

```
1 0
1 1
```

✓ Mandatory Test Cases

Test Case 1

KEYWORD

```
while(true)
```

✓ Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

4

Test Case 2

TOKEN COUNT

535

Test Case 3

NLOC

95